

Premise - Optical Characteristics 900 µm TB

This specification covers fibre characteristics for Single-mode Fibre ITU-T Rec. G.652.D and G.657.A1 and Multimode fibre OM1, OM3 and OM4 up jacketed to 900 µm in designs manufactured at the AFL Australia facility.

SINGLE-MODE FIBRE		
FIBRE SPECIFICATION (FIBRE CODE DESIGNATOR)		G.657.A1 (1F)
STRUCTURAL		
Typical Mode Field Diameter	@1310 nm	9.2 ± 0.4 µm
	@1550 nm	10.4 ± 0.5 µm
Cladding Diameter		125 ± 0.7 µm
Max Mode Field Concentricity Error		0.5 µm
Cladding Non-circularity		≤ 0.7%
Fibre Coating Diameter (Uncoloured)		240 ± 5 µm
Tight Buffered Diameter		900 +/- 0.05 µm
OPTICAL (NAT. FIBRE CHARACTERISTICS ONLY)		
Max. Attenuation (Cabled)	@1310 nm	0.4 dB/km
OS1/OS2 Compliant	@1550 nm	0.3 dB/km
Group Refractive Index	@1310 nm	1.4675
	@1550 nm	1.4681
	@1625 nm	1.4685
100 Turns, 50 mm Diameter	@1310 nm	≤ 0.01 dB
	@1550 nm	≤ 0.01 dB
10 Turns, 30 mm Diameter	@1550 nm	≤ 0.05 dB
	@1625 nm	≤ 0.3 dB
1 Turn, 20 mm Diameter	@1550 nm	≤ 0.5 dB
	@1625 nm	≤ 1.5 dB
Max Cut-off Wavelength (λ _{cc})		1260 nm
Zero Disp ⁿ Wavelength		1302 - 1324 nm
Slope at ZDW		0.073≤ S ₀ ≤0.092 (ps/nm².km)
Typ Ch. Disp Coefficient	@1285 – 1330 nm	≤ 3.5 ps/(nm.km)
	@1550 nm (D ₁₅₅₀)	13.3 ≤ D ₁₅₅₀ ≤ 18 ps/(nm.km)
	@1625 nm (D ₁₆₂₅)	17.2 ≤ D ₁₆₂₅ ≤ 22 ps/(nm.km)
PMD Ind.Fibre/LDV		≤0.1/≤0.04 (ps/√km)
MECHANICAL		
Proof Test		≥ 1 %

FIBRE IDENTIFICATION	
Fibre Colour Sequence (1st 12f)	Blue, Orange, Green, Brown, Grey, White, Red, Black, Yellow, Violet, Pink, Aqua
Fibre Colour Sequence (2nd 12f)	Blue*, Orange*, Green*, Brown*, Grey*, White*, Red*, Black*, Yellow*, Violet*, Pink*, Aqua* * With band mark

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MULTIMODE FIBRE				
FIBRE SPECIFICATION (FIBRE CODE DESIGNATOR)		OM1 - 62.5 µm (62)	OM3 - 50 µm (53)	OM4 - 50 µm (55)
STRUCTURAL				
Typical Core Diameter		62.5 ± 2.5 µm	50.0 ± 2.5 µm	50.0 ± 2.5 µm
Max Core-Clad Conc Error		1 µm	1 µm	1 µm
Cladding Diameter		125 ± 1 µm	125 ± 1 µm	125 ± 1 µm
Fibre Coating Diameter (Natural)		242 ± 5 µm	242 ± 5 µm	242 ± 5 µm
Tight Buffered Diameter		900 +/- 0.05 µm	900 +/- 0.05 µm	900 +/- 0.05 µm
MECHANICAL				
Proof Test		1 %	1 %	1 %
OPTICAL				
Min. G-Ethernet Transmission Distance	@850 nm	300 m	1000 m	1040 m
	@1300 nm	500 m	550 m	550 m
Max. Attenuation (Cabled)	@850 nm	3.0 dB/km	3.0 dB/km	3.0 dB/km
	@1300 nm	1.0 dB/km	1.0 dB/km	1.0 dB/km
Min. Overfilled Bandwidth	@850 nm	200 MHz.km	1500 MHz.km (300 MHz.km 10GigEthernet)	3500 MHz.km (500 MHz.km 10GigEthernet)
	@1300 nm	500 MHz.km	500 MHz.km	500 MHz.km
Numerical Aperture		0.275 ± 0.015	0.200 ± 0.015	0.200 ± 0.015

FIBRE IDENTIFICATION	
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Fibre Colour Sequence (2nd 12f)	Blue*, Orange*, Green*, Brown*, Grey*, White*, Red*, Black*, Yellow*, Violet*, Pink*, Aqua* * With band mark

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